Investigation of Associations Between Donor Risk Index Variables and Delayed Liver Graft Function

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Introduction

Methods

Results

Logistic Regression Model

$$
\begin{aligned}
\hat{y}_i = & -2.106469 - 0.005237(\text{donor age}_i) \\
& - 0.312951(\text{donor cause of death}_i = \text{head trauma}) \\
& - 0.255460(\text{donor cause of death}_i = \text{anoxia}) \\
& + 0.763904(\text{donor cause of death}_i = \text{other}) \\
& - 0.013151(\text{donor race}_i = \text{caucasian}) \\
& - 0.094485(\text{donor sex}_i = \text{female}) \\
& + 1.046880(\text{donor risk index}_i)
\end{aligned}
$$

$\hat{y}_i$ represents the predicted log-odds of delayed liver graft function.

Table 1: Logistic Regression Model Coefficients

Donor.Variable	Estimate	Odds.Ratio	P.Value
Age	-0.005	0.995	0.528
Cause of Death: Head Trauma	-0.313	0.731	0.300
Cause of Death: Anoxia	-0.255	0.775	0.432
Cause of Death: Other	0.764	2.147	0.177
Caucasian	-0.013	0.987	0.962

Donor.Variable	Estimate	Odds.Ratio	P.Value
Female	-0.094	0.910	0.703
Donor Risk Index	1.047	2.849	0.010

Interpretation:

The intercept term -2.106469, representing the log odds of a recipient experiencing delayed liver graft function from a non-white male donor of age zero who's cause of death was cerebrovascular accident, is not a relevant real-world scenario.

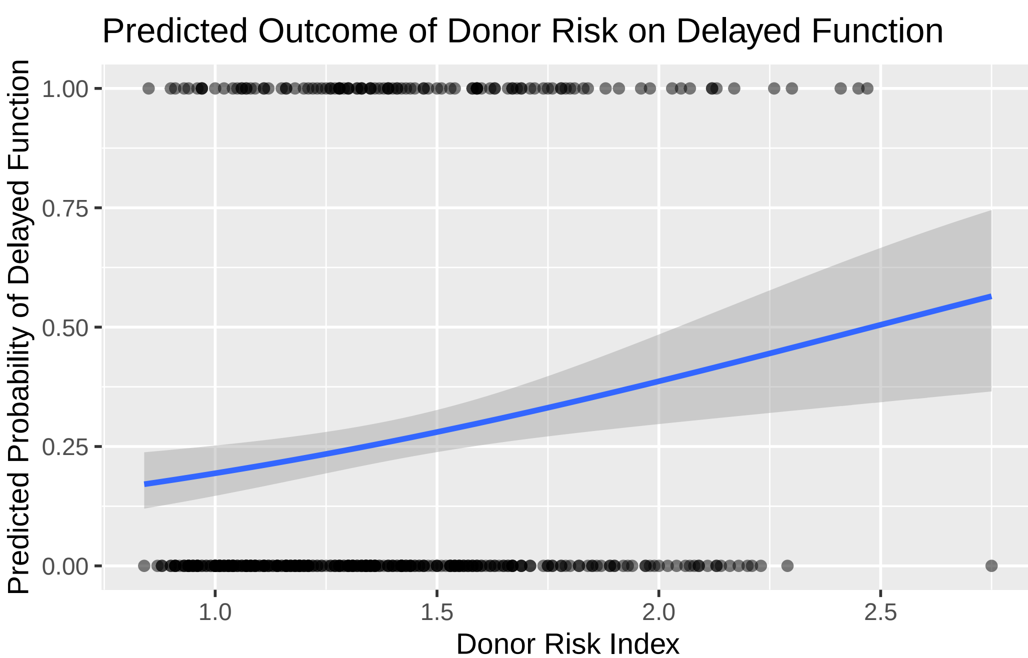
Donor Age: This model predicts a recipients log odds of having delayed liver graft function to decrease by 0.005237 given a one year increase in age of their donor, while controlling for all other variables. In other words, the recipient is predicted to have 0.9947767 times the odds of experiencing delayed liver graft function given a one year increase in age of their donor, while controlling for all other variables. The corresponding p-value for this coefficient, the probability of seeing this predicted change in log odds for a one year change in age given that the true population change in log odds is zero, is 0.528224. Evaluated against the significance level of 0.05, we fail to reject the null hypothesis that the true slope value is zero. We do not have sufficient statistical evidence to support there being an association between donor age and delayed liver graft function.

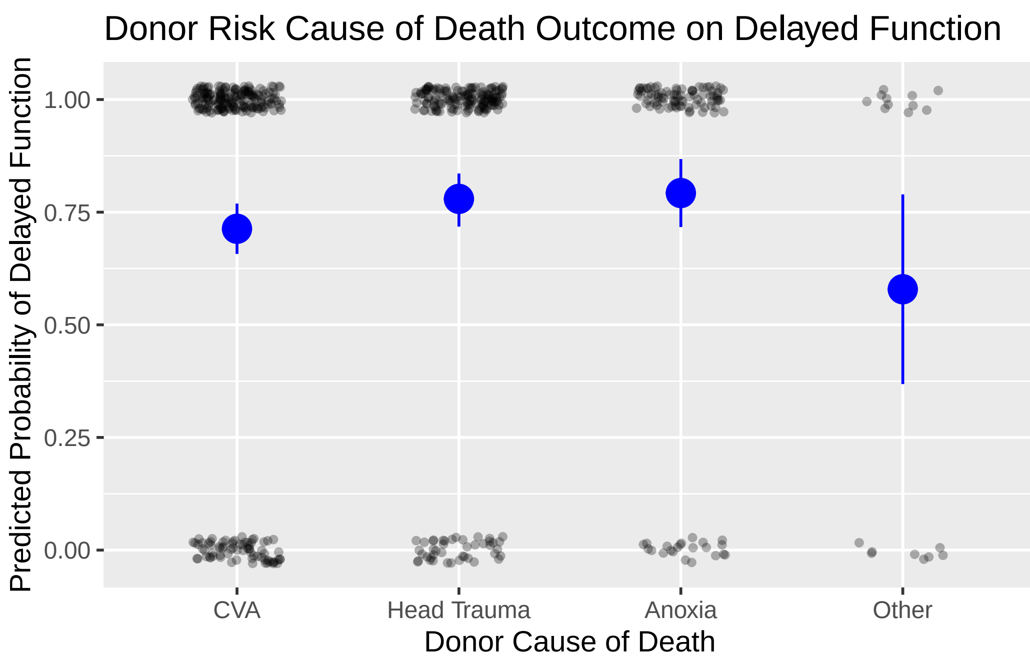
Donor Cause of Death: This model predicts a recipients log odds of having delayed liver graft function to change by -0.312951 if the donor's cause of death was head trauma, -0.255460 if the donor's cause of death was anoxia, and 0.763904 if the donor died of some other cause, compared to a recipient who's donor died of cerebrovascular accident, while controlling for all other variables. The recipient is predicted to have 0.7312857, 0.7745601, or 2.14664 times the odds of experiencing delayed liver graft function given the donor cause of death was head trauma, anoxia, or other, respectively. The corresponding p-value for this coefficient, the probability of seeing this predicted change in log odds given that donor cause of death does not truly change the log odds, are 0.528224, 0.300045, and 0.176776, respectively. Evaluated against the significance level of 0.05, we fail to reject the null hypothesis that the true slope value is zero. We do not have sufficient statistical evidence to support there being an association between donor cause of death and delayed liver graft function.

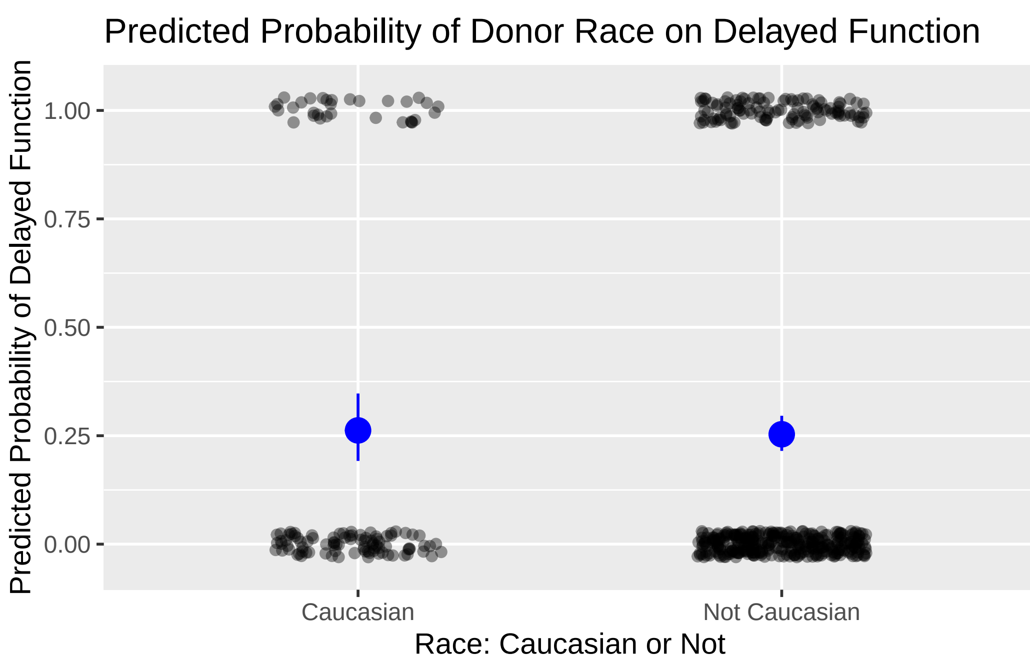
Donor Whiteness: This model predicts a recipients log odds of having delayed liver graft function to be 0.013151 lower for patients with caucasian donors compared to patients with non-caucasian donors, while controlling for all other variables. In other words, the recipient is predicted to have 0.9869351 times the odds of experiencing delayed liver graft function if their donor is caucasian versus non-caucasian, while controlling for all other variables. The corresponding p-value for this coefficient, the probability of seeing this predicted change in log odds of delayed liver graft function for caucasian donors given that the true population difference in log odds is zero, is 0.961797. Evaluated against the significance level of 0.05, we fail to reject the null hypothesis that the true slope value is zero. We do not have sufficient statistical evidence to support there being an association between donor gender and delayed liver graft function.

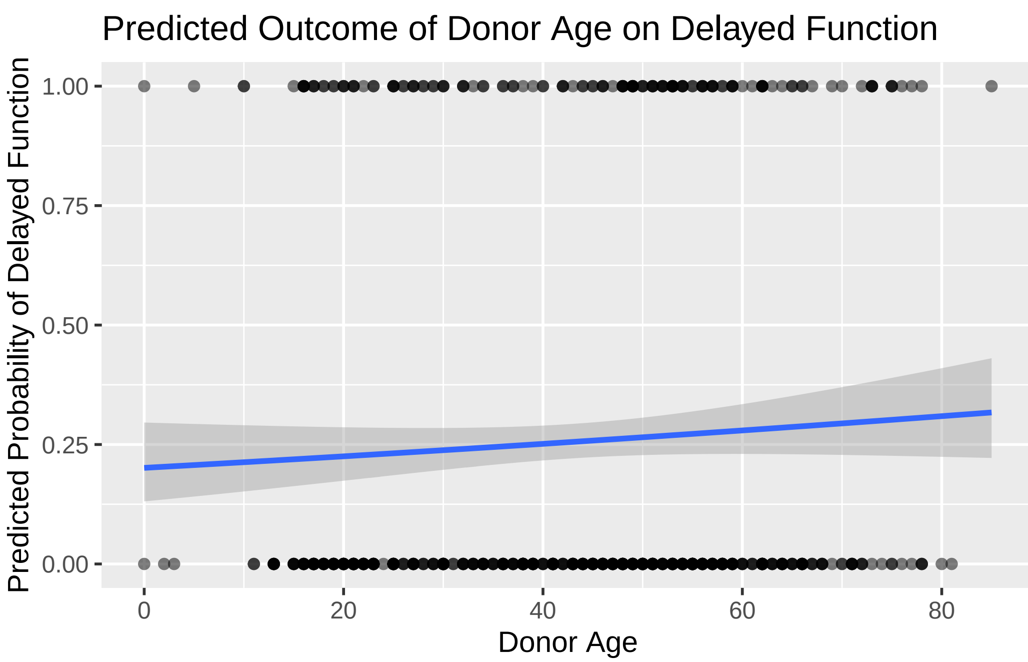
Donor Gender: This model predicts a recipients log odds of having delayed liver graft function to be 0.094485 lower for patients with female donors compared to patients with male donors, while controlling for all other variables. In other words, the recipient is predicted to have 0.9098414 times the odds of experiencing delayed liver graft function if their donor is female versus male, while controlling for all other variables. The corresponding p-value for this coefficient, the probability of seeing this predicted change in log odds of delayed liver graft function for female donors given that the true population difference in log odds is zero, is 0.702822. Evaluated against the significance level of 0.05, we fail to reject the null hypothesis that the true slope value is zero. We do not have sufficient statistical evidence to support there being an association between donor whiteness and delayed liver graft function.

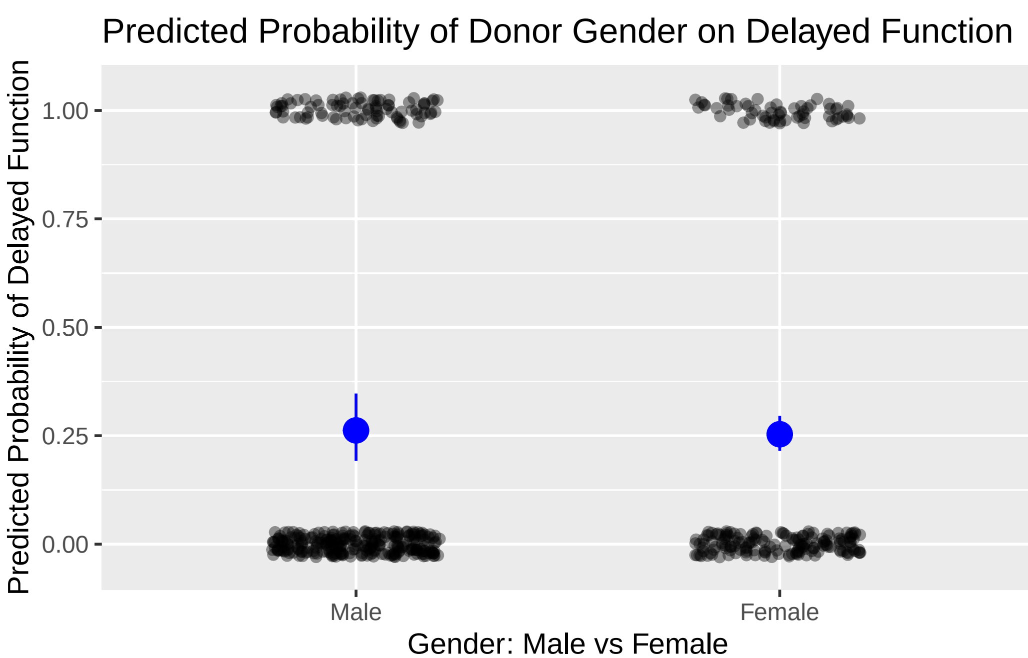
Donor Risk Index: This model predicts a recipients log odds of having delayed liver graft function to be 1.046880 higher given a one unit change in donor risk index, while controlling for all other variables. In other words, the recipient is predicted to have 2.848749 times the odds of experiencing delayed liver graft function given a one unit change in donor risk index, while controlling for all other variables. The corresponding p-value for this coefficient, the probability of seeing this predicted change in log odds of delayed liver graft function given that the true population slope in log odds is zero, is 0.009538. Evaluated against the significance level of 0.05, we reject the null hypothesis that the true slope value is zero. There is sufficient statistical evidence to support there being an association between donor risk index and delayed liver graft function.




Donor Risk Cause of Death Outcome on Delayed Function
Predicted Probability of Delayed Function
1.00
0.75
0.50
0.25
0.00
CVA
Head Trauma
Anoxia
Other
Donor Cause of Death


Predicted Probability of Donor Race on Delayed Function
Predicted Probability of Delayed Function
1.00
0.75
0.50
0.25
0.00
Caucasian
Not Caucasian
Race: Caucasian or Not


Predicted Outcome of Donor Age on Delayed Function
Predicted Probability of Delayed Function
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
Donor Age


Predicted Probability of Donor Gender on Delayed Function
Predicted Probability of Delayed Function
1.00
0.75
0.50
0.25
0.00
Male
Female
Gender: Male vs Female

Discussion

Our analysis shows that out of all of the liver donor characteristic variables we tested for, we only found evidence of one statistically significant relationship, that being the one between donor risk index and delayed liver graft function. This is in spite of the fact that all of the other characteristics we tested for are varaibles used in the calculation of the donor risk index. The p-values associated with those variables, which were donor age, cause of death, race, and gender, were above our significance level of 0.05. However, this may be because some of those variables, such as gender and race, do not seem to have some obvious, direct relationship with liver function. This is especially apparent in the fact that our variable for race was binary, with the only categories being "Caucasian" and "not Caucasian," which could be considered biased and reductive. Overall, the scope of our tests may have been limited by our lack of other donor-related variables. Our testing could have been strengthened by having other donor variables, specifically those that are used in calculating the donor risk index, such as history of diabetes, history of hypertension, and donation after circulatory death (DCD) status. Nonetheless, without being able to test for all of the variables used to calculate DRI, our ability to come to conclusions about which of those variables have statistically significant relationships with delayed liver graft function is extremely limited.